

Xuan Huang

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🌐 [Personal Website](#)

Educational Background

University of Electronic Science and Technology of China

Sept. 2022 - Jun. 2026 (expected)

B.Eng in Communication Engineering

Rank – 4/189 (top 2.1%) Average Score – 94.2/100 GPA – 3.93/4

Shanghai Jiao Tong University

Sept. 2026 - Jun. 2031 (expected)

Ph.D. in Electronic and Information Engineering

Research Experiences

Efficient Token Compression for Multimodal Large Language Models

Dec. 2025 – Present

Undergraduate Thesis Project & Research Intern, Ant Group

- **Streaming Token Redundancy Modeling:** Analyzed temporal redundancy and semantic repetition in streaming video tokens; developed a redundancy identification framework tailored for streaming scenarios.
- **Token Compression Algorithm Design:** Designed a pre-LLM token pruning strategy that preserves causal reasoning, leveraging semantic segmentation and cross-frame feature similarity to retain key tokens while reducing input size.
- **Streaming Inference Framework Implementation:** Built an experimental pipeline based on pretrained VLMs, integrating token compression with KV-cache-based streaming inference; evaluated compression ratio, latency, and task performance.

Self-powered Multimodal Emotion Recognition System

Dec. 2023 – Aug. 2024

National Undergraduate Innovation and Entrepreneurship Program (Excellent Project)

- **Multimodal Fusion Model:** Built a dual-modal model with modality-specific/invariant feature fusion for speech and wavelet-based EEG preprocessing for ViT input.
- **IoT System Development:** Developed full-stack system with ESP32-based EEG acquisition, Huawei Cloud for IoT data storage, Alibaba Cloud for backend model deployment, and WeChat Mini Program frontend.
- Published a **First-author** paper which is accepted by *ICDT 2025* (see Section “Publications”).
- **National First Prize** in Huawei IoT Design Competition (Top 3.6%).

Publications

[C1] **X. Huang**, Y. Zhang, X. Zhang, X. Zhang, and H. Tang, “Voice Recognition System for Speech-to-Text and GPT Communication Powered by Organic Photovoltaic,” accepted by *ICDT 2025*.

[C2] X. Wei, W. Mei, **X. Huang**, Z. Chen, and B. Ning, “Mechanical Power Modeling and Energy Efficiency Maximization for Movable Antenna Systems,” accepted by *IEEE GLOBECOM 2025*[PDF].

[J1] X. Wei, W. Mei, **X. Huang**, Z. Chen, and B. Ning, “Energy-Efficient Movable Antennas: Mechanical Power Modeling and Performance Optimization,” submitted to *IEEE TWC*[PDF].

Honors and Awards

Outstanding Graduate of Sichuan Province (Top 4%)

2025 Tencent First-Class Scholarship (“Chengdian Friend Scholarship”)

2024 **National Scholarship (Top 1.5%)** for Undergraduates

2023 & 2024 & 2025 UESTC Outstanding Student Scholarship (Top 8%)

2024 National IoT Design Competition

National First Prize

2024 Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)

National Second Prize

2024 National English Competition for College Students (NECCS)

National Second Prize

2024 Mathematical Contest in Modeling (MCM)

Honorable Mention

Skills and Services

Technical skills: Experienced in **Python**, **C/C++** and **Matlab**. Available for **JavaScript** and **Verilog**.

Language: Chinese (native) English: CET-4: **664** CET-6: **625**

Social services: Certified Peer Counselor at UESTC, Class Study Committee Member (“Excellent Class”).